Heat pipes and nanofluids utilization for cooling photovoltaic panels: an application of hybrid machine learning and optimization models

Aysar A. Abdulrahman*,†
Department of Computer Networks, College of Informatics, Sulaimani Polytechnic University, Sulaymaniyah, Iraq

Abstract

This study explores the synergies between advanced cooling technologies and photovoltaic systems, seeking to improve their overall efficiency and contribute to the broader goal of mitigating greenhouse gas emissions. To cool photovoltaic panels in more efficiently maner, understanding heat pipes, nanofluids, and panels interaction play key roles. For analysis and optimization, hybrid models of convolutional neural network (CNN) and firefly optimization algorithm are employed. The firefly optimization algorithm is used to optimize the thermosiphon heat pipe's operational conditions, taking into account inputs such as the filling ratio, nanofluid concentration and panel angle. The study compared the predicted outcomes of a classic CNN model to laboratory experiments. While the CNN model was consistent with experimental findings, it struggled to predict high power values with precision. The proposed model improved high power value predictions by 4.05 W root mean square error (RMSE). The proposed model outperformed the classic CNN model for values greater than 50 W, with an RMSE of 3.95 W. The optimal values for the filling ratio, nanofluid concentration and panel angle were determined after optimization with the firefly algorithm. The research contributes to the advancement of renewable energy technologies and the optimization of photovoltaic panel cooling and energy production. Nanofluid with 1% mass concentration improves photovoltaic collector thermal efficiency due to its higher thermal conductivity coefficient. The photovoltaic collector's electrical efficiency peaks in the morning, drops at noon due to temperature and radiation and recovers by morning. Electrical efficiency is best with nanofluid at 0.86%. Exergy efficiency closely matches electrical efficiency, with nanofluid at the optimal percentage achieving the highest efficiency and water cooling the lowest.

Keywords: photovoltaic panel; nanofluids; heat pipe; convolutional neural network; firefly optimization algorithm; photovoltaic panel efficiency

*Corresponding author.
aysser.abdulrahman@spu.edu.iq,
aysser.abdulrahman@gmail.comReceived 21 November 2023; revised 29 January 2024; accepted 15 March 2024

1. Introduction

In recent years, the urgency of addressing climate change has underscored the need for sustainable and eco-friendly practices in various technological domains [1–3]. One such critical area is the field of renewable energy, where the deployment of photovoltaic panels has become increasingly prevalent as a clean and efficient means of electricity generation [4–8]. While the utilization of solar energy through photovoltaic panels offers a promising avenue for reducing reliance on conventional fossil fuels, optimizing the efficiency of these panels is essential for maximizing their environmental benefits, especially low carbon emission [9–11].

Various liquids, including water, oil and ethylene glycol, are of significant importance in numerous engineering applications,

†, https://orcid.org/(0000–0001–9589-0605)
International Journal of Low-Carbon Technologies 2024, 19, 1078–1088
© The Author(s) 2024. Published by Oxford University Press.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.
https://doi.org/10.1093/ijlct/ctae047

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

such as casting procedures, heat exchange mechanisms and cooling systems for electronic equipment [12–14]. Various techniques have been employed in order to enhance the heat transfer coefficient of these liquids owing to their relatively low coefficient of thermal conductivity [15–17]. One method for enhancing the heat transfer coefficient of these liquids involves the use of solid nanoparticles, a technique initially introduced by Choi and Eastman [18]. Numerous scientific investigations have been undertaken by researchers to quantify the thermal conductivity coefficient of nanofluids [19–25]. The thermal conductivity coefficient of nanofluids is influenced by various factors such as temperature, nanoparticle shape and size, volume percentage of nanoparticles and the type of base fluid [26–28].

Solar heating systems are designed to transform the radiant energy emitted by the sun into thermal energy. Photovoltaic panels, comprising solar cells, serve as the primary component of photovoltaic systems, facilitating the conversion of solar radiation into electrical energy [29–33]. The elevated temperature of the solar cells results in a reduction in the electrical efficiency. Studies have demonstrated that the application of cooling techniques to photovoltaic panels can effectively lower the temperature of solar cells. Consequently, this cooling process enables photovoltaic panels to provide electricity that closely aligns with their specified nominal power output. Hence, the thermal management of solar panels holds significant relevance. The major solar panels have been cooled through the utilization of air, water and heat pipes [34–37].

Researchers have utilized soft computing methods, to predict the thermal conductivity of nanofluids. This approach has been made possible due to the advancements in computer science and software. Papari *et al.* [38] employed a neural network to estimate the thermal conductivity coefficient of several nanofluids. In their study, Hojjat *et al.* [39] conducted measurements of the thermal conductivity coefficient of several nanofluids comprising nanoparticles of aluminum oxide, titanium dioxide and copper oxide. These measurements were carried out using laboratory-based methodologies. The application of neural network methodology was employed to simulate the outcomes of the laboratory experimentation. The input variables under consideration were the temperature and volume fraction of nanoparticles, while the output variable of interest was the thermal conductivity coefficient of the nanofluid. Longo *et al.* [40] employed two neural networks in order to ascertain the coefficient. The thermal conductivity of nanofluids comprising aluminum oxide and titanium dioxide was suggested. The independent variables considered in this study were the temperature, volume percent and diameter of nanoparticles. Hemmat Esfe *et al.* [41] and Afrand *et al.* [42] have employed neural network models and conducted empirical investigations to establish predictive associations.

The cooling of photovoltaic panels is a pivotal aspect of enhancing their performance and longevity, and this is where the innovative integration of heat pipes and nanofluids comes into play [43]. By implementing hybrid machine learning and optimization models, it can be aimed to develop a comprehensive understanding of how heat pipes and nanofluids can be tailored to enhance the cooling mechanisms of photovoltaic panels [44]. The reduction of heat-induced inefficiencies in photovoltaic panels not only enhances energy output but also contributes to a lower carbon footprint by promoting the sustainable utilization of solar power [45].

An extensive analysis of previous research has shown that using a digital computational tool to demonstrate the experimental correlation and modeling is a highly effective approach that has generated considerable interest among many scholars. These methodologies have the capacity to replace repetitive examinations, which can be both laborious and costly. However, the evaluation of separate and independent factors has not been thoroughly examined due to the increasing costs of computation and the complex process of assessing networks related to the power of the photovoltaic system. Therefore, this study utilized a combined methodology that incorporated a convolutional neural network (CNN) and firefly optimization technique to evaluate the influence of nanofluid and heat pipes on the efficiency of the photovoltaic system. The study has employed laboratory data as the input for the machine learning model. In particular, the model was developed using a total of 100 laboratory samples. This study supports the worldwide effort to address climate change and emphasizes the importance of incorporating advanced technologies to enhance renewable energy systems for a more sustainable and environmentally aware future.

2. Materials and methods

Figure 1a depicts the system used in the current study, which cools the system by flowing nanofluid dissolved in water behind the tubes of the photovoltaic panel. Figure 1b illustrates the primary components of a cylindrical heat pipe.

In this study, a small thermal photovoltaic panel measuring 0.24 m^2 was used. To measure radiation intensity from an SPM-1116 SD radiation meter with an accuracy of 0.1 watt per square meter and a measurement range of 0–2000 W/m^2, an ultrasonic device with a KS250F probe model with a frequency of 20 kHz and power of 250 W, analyzer. The PROVA 200A model was used to obtain the circuit voltage and short circuit current with a voltage determination accuracy of 1%, in the range of 0 to 60 V, and a current determination accuracy of 1%, in the range of 0 to 6 A. WANT WT5003G model was used scale for Measurement the required amount of nanoparticles with 500g/1mg accuracy. An YF-S201 model flow meter made by Sea Company in China with a working range of 1–30 lit/min was used. The experiments were conducted during the summer season in the Sulaymaniyah, Iraq located at 45.41 degrees longitude and 35.56 degrees latitude. The solar meter displays the radiation intensity associated with the photovoltaic/heat collector, as depicted in Figure 2.

It is notable that the majority of the presented simulation and optimization algorithms are computer simulations of natural processes. In this section, the structure and operation of the CNN and firefly optimization algorithm used in this study are described.

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

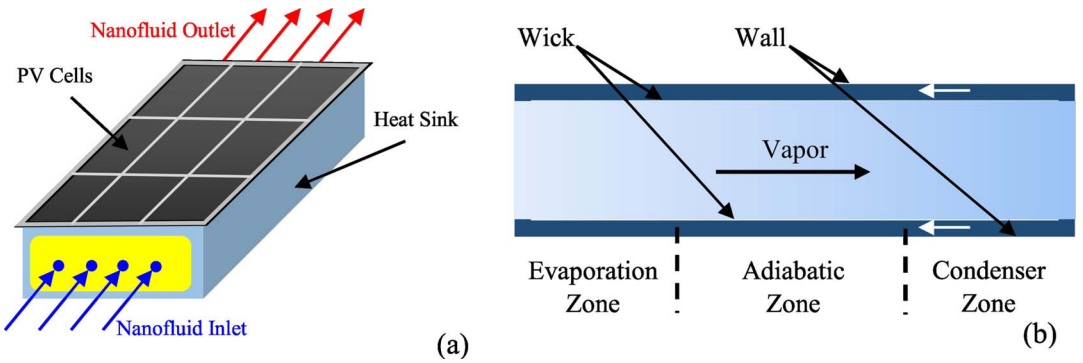


Figure 1. *A schematic of (a) photovoltaic cells cooling system with nanofluids (b) a heat pipe.*

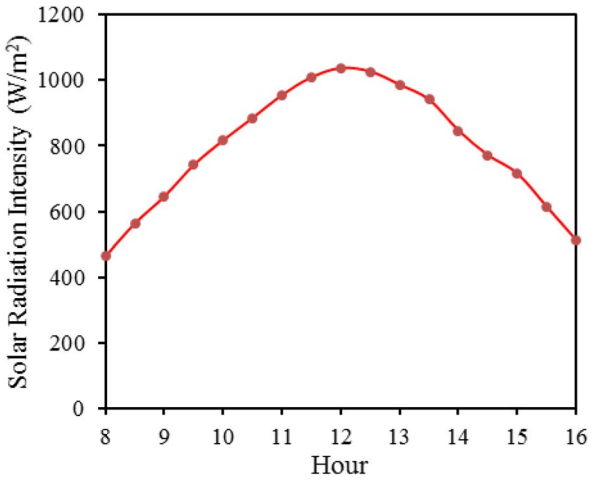


Figure 2. *Solar radiation intensity related to the studied area.*

The thermal and electrical efficiency of the solar panel is assessed by testing it with various concentrations of nanofluids and water.

2.1. CNN

Pattern recognition refers to the ability of computers to discern distinctions among various patterns. It is intricately linked to machine learning and artificial intelligence and finds extensive use in the domain of machine learning across several fields [46–48].

The machine learning techniques under consideration possess a topology like that of a network, enabling the effective extraction of significant characteristics and information [49, 50]. This is achieved by sequentially traversing a set of convolutional layers, each of which is characterized by a specific kernel size [51]. The neuron-based model encompasses a multitude of training weights and biases, serving the purpose of extracting, classifying and predicting data. At the beginning of training, these weights and biases are assigned in a random manner. The convolutional layer, an essential component of CNNs, performs regular calculations of the inner product between the filters and the input data. The input factor plays a crucial role [52].

The convolution operation may be stated as the algebraic operation associated with the internal multiplication between the values of each filter and the corresponding input data. A filter, also known as a kernel, is a matrix that possesses a predetermined size and shares the same depth as the incoming data. Additionally, it is important to acknowledge that the initial convolutional layers are tasked with extracting low-level characteristics, whilst the subsequent convolutional layer is responsible for detecting higher level features [53].

The utilization of batch normalization proves to be advantageous in scenarios involving deep layers, as it facilitates the scaling and standardization of the output from each layer. This process is performed for the input of the subsequent layer and is executed on a per mini-batch basis. The process of standardization involves recalculating the data to get the standard deviation and mean of zero of one [54].

Since the effects of vanishing gradients are problematic, a desired activation function based on our input data is applied on each feature to improve the computational efficiency by reducing the dispersion. Usually, the output of the convolution layer or normalization layer is fed to an activation layer. In this study, the rectified linear unit (ReLU) activation function [55], which is based on the backpropagation of errors, is applied. Equation (1) defines the ReLU activation function.

$$y = \max(0, x), \tag{1}$$

where x is the input of the ReLU activation function and y is its output.

A fully connected (FC) layer has also been employed to extract robust high-level features from the input data. Inside this particular stratum, every individual node is subjected to a multiplication operation with its corresponding adjustable weight value inside each input vector [56]. In order to utilize the output of the convolution layer or the activation layer in the subsequent FC layer, it is necessary to transform all the features into a vertical vector.

In contrast to convolutional layers, FC layers include a higher number of parameters, hence rendering them more challenging to train. The last component of the CNN design is the regression layer. This layer is responsible for capturing the connection

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

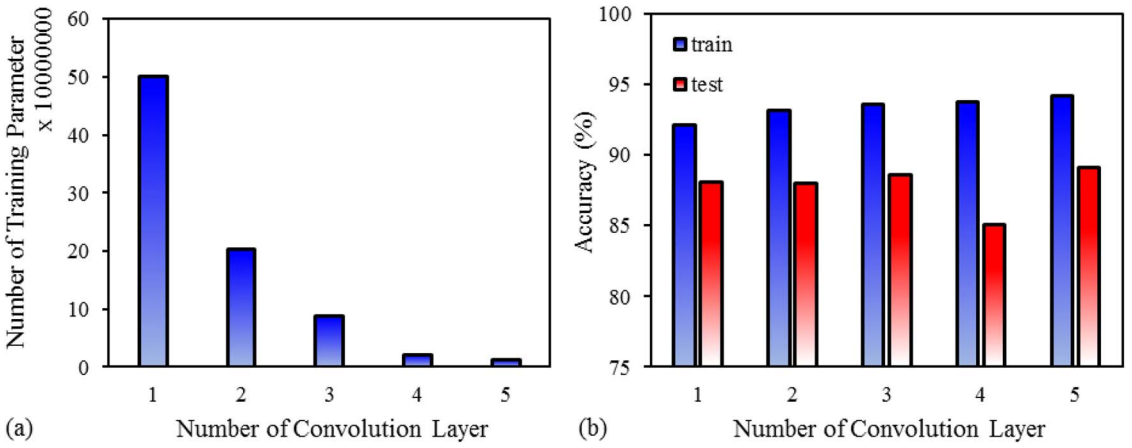


Figure 3. *(a) The number of parameters of each training model (b) accuracy of models.*

between a desired outcome and one or more independent variables, also known as input data [57].

The process of training a CNN involves iteratively adjusting the weights and biases in each layer of the model to minimize a predefined cost function. The iterative process of reducing a cost function involves the utilization of a gradient descent approach, which includes the computation of the derivatives of the cost function [58].

CNN models have the ability to extract new and unique information from an input matrix. The experiments aimed to optimize the output power of the photovoltaic panel by adjusting various parameters. The parameters included the solar panel's angle, the heat pipe's filling ratio and the nanofluid concentration. The experiments were carried out under controlled conditions, with the ambient air temperature and solar radiation intensity remaining constant. The goal was to find the combination of parameters that would result in the maximum output power of the photovoltaic panel.

There are nine layers of convolution with varying numbers of filters. Three 2×1 filters have been applied to the initial three layers. The output of the normalization layer of groups is activated with the ReLU activation function to improve training and normalize the data values. The layer that follows contains 16 filters for feature extraction, followed by three layers containing 32 related filters. Once more, the bulk normalization layer and activation layer are utilized. Four layers of manual normalization and the ReLU function are used to maintain the output values of the layers within the normal range and prevent overfitting. In the final convolution layers, two layers with 128 filters are followed by two layers with no filters. In addition, two dropout layers are used to reduce the effect of overfitting by controlling the fitting process. In the proposed model, it is unnecessary for the data to be back-to-back. Each model's number of training parameters and the precision of its training and evaluation data are specified in Figure 3.

The proposed convolution structure to minimize the loss in Equation (2) uses the root mean square error (RMSE) index, which calculates the difference between the actual and estimated (predicted) values.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\text{estimated} - \text{actual})^2}, \tag{2}$$

where n specifies the predicted number or the actual value.

2.2. Optimization algorithm

In the firefly algorithm, fireflies move randomly in space, and the amount of light emitted increases the firefly's attractiveness. However, as distance increases, the amount of this luminosity decreases, and the firefly's attractiveness decreases. The intensity of light at distance r is given by the expression in Equation (3). The level of brightness and attractiveness is one of the characteristics of fireflies in the firefly algorithm. The amount of light intensity at the distance r of a light source of points whose light intensity is I_0 is equal to I_0/r^2 [59].

$$I = I_0 e^{-\gamma r^2} \approx \frac{I_0}{1+\gamma r^2}. \tag{3}$$

The firefly is considered as a point-light source in the algorithm. By combining the light absorption and light attenuation formulas, Equation (3) will be obtained and Equation (4) will be used to show the level of attractiveness (β) in general [60].

$$\beta = \beta_0 e^{-\gamma r^m} \approx \frac{\beta_0}{1+\gamma r^m}, \tag{4}$$

where m is a non-negative number and $m = 0$ means that the amount of attractiveness is the same at any distance, and the higher the value of m is, it indicates that with the increase of the distance, the amount of attractiveness decreases from night to day. It gets cooler. In other words, the attractiveness does not decrease

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

with increasing distance (m never becomes zero and its value is always considered greater than one). In the above formulas, γ is used and because γ is a function of the spatial position and is the reciprocal of the square of the distance. Therefore, it is recommended to use the parameter Γ which $\Gamma = \frac{1}{\sqrt{\gamma}}$. Because this parameter is no longer dependent on the spatial position and is of the distance type.

Therefore, the new position of the firefly, x, whose previous position was x, and which is being attracted to the firefly with a greater amount of light at position x, is calculated using Equation (5), where e is the amount of light emitted by the firefly at position x. It is a vector with a Gaussian or uniform distribution. γ is the light absorption coefficient, is the mutation coefficient and its value can be changed (decreased) in each iteration until the algorithm converges; and this change can be linear or exponential [61].

$$x_i' = x_i + \beta_0 e^{-\gamma r^m}\left(x_j - x_i\right) + \alpha\varepsilon_i, \tag{5}$$

To optimize the operational conditions, a cost function must first be defined in the optimization algorithm. In this study, a properly trained neural network was used to define the optimization function. In conclusion, the optimization function incorporates five input variables, which are subsequently utilized in conjunction with the optimal algorithm. The primary objective of this optimization algorithm is to maximize the output power generated by the photovoltaic panel.

2.3. Heat pipes

Heat pipes are widely recognized as highly efficient heat transfer systems and are considered to be among the most well-known systems in this category. One notable benefit associated with the utilization of heat pipes is their ability to efficiently transfer heat over long distances, without necessitating an external driving force to facilitate circulation. Heat pipes are thermal transfer devices that operate in two phases and possess remarkably high effective thermal conductivity. These devices can be designed with either cylindrical or flat structures [62]. Heat pipes are composed of three primary components: the evaporator section, the adiabatic section and the condenser section. The movement of steam from the evaporating section to the condensing section in heat pipes is driven by the pressure gradient. Simultaneously, the liquid is transported from the condensing region to the evaporating region through the exertion of various forces, including capillary force, gravity force, electrostatic force and others.

Machine learning methods significantly increase the speed of calculations by simulating data obtained from experiments while maintaining accuracy and precision. In order to achieve the optimal weighting coefficients of the CNN for the best prediction of the operating conditions, the optimization algorithm has been used.

To model this process, the input parameters of solar radiation intensity, air temperature, solar panel angle, filling percentage and nanofluid concentration are used as model inputs, with photovoltaic panel output power as the output parameter. Laboratory samples have been given a value of 100. Of the sampling points, 80 were used for model training and 20 for model testing.

Optimum determination of the number of neural network layers, the number of neurons and the transfer function in each layer and the learning function used in the network is very important. In fact, it is tried to create a network with the least error and make the most generalizations to new and unknown data.

2.4. Energy and exergy efficiency

This study initially examines the photovoltaic collector with and without a cooling system. Subsequently, the laboratory assessment focuses on the impact of altering the flow rate and mass concentration of nanofluid on the system's electrical and thermal efficiency.

2.4.1. *Thermal efficiency*

Thermal efficiency is a metric employed in the thermal analysis of a thermal photovoltaic collector. It is determined using Equation (6).

$$\eta_{\text{th}} = \frac{\dot{Q}_{\text{u}}}{\dot{Q}_{\text{s}}} \times 100, \tag{6}$$

where $\dot{Q}_{\text{u}}$is the heat rate absorbed by the working fluid flowing in the ducts and $\dot{Q}_{\text{s}}$ is the rate of radiant energy hitting the panel. The values of $\dot{Q}_{\text{u}}$and $\dot{Q}_{\text{s}}$ are calculated from Equations (7) and (8), respectively.

$$\dot{Q}_u = \dot{m}C_P\Delta T = \dot{m}C_P\left(T_{\text{out}} - T_{\text{in}}\right) \tag{7}$$

$$\dot{Q}_s = A_c I(t), \tag{8}$$

In Equation (8), $I(t)$ and A_{c} are the intensity of solar radiation and the surface area of the collector, respectively.

2.4.2. *Electrical efficiency*

Electrical efficiency is a frequently used measure to describe the electrical performance of a thermal photovoltaic collector. It is defined as the ratio of the actual electrical power output from the photovoltaic module to the energy of solar radiation that reaches the surface of the module. This value is calculated using Equation (9).

$$\eta_{\text{el}} = \frac{P_{\text{mp}}}{A_c I(t)} \times 100, \tag{9}$$

P_{mp}represents the highest amount of electric power that the panel can generate and is determined using Equation (10).

$$P_{\text{mp}} = V_{\text{mp}} I_{\text{mp}} - P_{\text{pump}}. \tag{10}$$

2.4.3. *Exergy analysis*

Exergy, a crucial concept in the second law of thermodynamics, refers to the maximum achievable useful work that can be

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

obtained from a flow of matter or energy. The exergy efficiency of the thermal photovoltaic collector is calculated according to the net exergy rate of the system output with Equations (11) and (12).

$$\sum \mathrm{Ex}_{\mathrm{in}} + \sum \mathrm{Ex}_{\mathrm{out}} = \sum \mathrm{Ex}_{\mathrm{loss}} \tag{11}$$

$$\sum \mathrm{Ex}_{\mathrm{out}} = \sum \mathrm{Ex}_{\mathrm{termal}} + \sum \mathrm{Ex}_{\mathrm{electrical}}. \tag{12}$$

The input and output exergy are calculated from Equations (13)–(15).

$$\mathrm{Ex}_{\mathrm{in}} = A_{\mathrm{c}} \times I(t) \times \left[1 - \frac{4}{3} \times \left(\frac{T_{\mathrm{a}}}{T_{\mathrm{s}}}\right) + \frac{1}{3} \times \left(\frac{T_{\mathrm{a}}}{T_{\mathrm{s}}}\right)^4\right] \tag{13}$$

$$\dot{\mathrm{E}}\mathrm{x}_{\mathrm{thermal}} = \dot{\mathrm{Q}}_{\mathrm{u}} \left[1 - \frac{T_{\mathrm{a}} + 273}{T_{\mathrm{out}} + 273}\right] \tag{14}$$

$$\dot{\mathrm{E}}\mathrm{x}_{\mathrm{electrical}} = I_{\mathrm{mp}} \times V_{\mathrm{mp}}. \tag{15}$$

The exergy of the whole system is also obtained from Equation (16).

$$\eta_{\mathrm{ex}} = \frac{\mathrm{Ex}_{\mathrm{out}}}{\mathrm{Ex}_{\mathrm{in}}}. \tag{16}$$

3. Results and discussion

This section presents the outcomes of the CNN model pertaining to the simulation of heat pipe process behavior. Subsequently, the firefly algorithm is utilized to optimize neural convolution, followed by a comparison between the two proposed methods and the conventional approach.

3.1. Machine learning results

Figure 4a depicts the juxtaposition of the forecasted outcomes of a CNN and the actual values obtained from laboratory experiments. Based on the figure, the machine learning model's prediction outcomes, as measured by the RMSE index of 8.29 W, align with the experimental results. Furthermore, a close examination of the results reveals that this network is unable to accurately forecast the output pertaining to high values of photovoltaic panel output power. Figure 4b is a scatter plot displaying the dispersion of model and laboratory results. From Figure 4b, it is evident that the model lacks accuracy in estimating the power output of the photovoltaic panel, in large amounts.

Figure 5a displays the contrast between the predicted power of the solar panel using the suggested model and the actual experimental data. The RMSE index is reported to be 4.05 W. The figure clearly demonstrates the effectiveness of the suggested model in accurately anticipating high power values of the solar panel. This matter is observable in Figure 5b and the scatter plot depicting the experimental findings and model forecast. Thus, it appears that the combination of the suggested CNN model and the firefly optimization model can effectively address the limitation of accurately predicting high power values of the photovoltaic panel.

Table 1 demonstrates that the classical model's performance in forecasting the power of the photovoltaic panel is comparable to the proposed model for values under 50 W. Specifically, the RMSE for the classical model is 4.81 W, while for the suggested model it is 3.95 W. The suggested model demonstrates notably greater efficacy in predicting values upper than 50 W compared to the classic CNN model. For the CNN model, the value is 9.02 W, however for the proposed model, it is 4.17 W.

3.2. Optimization results

Due to the fact that the proposed model outperformed the conventional model, the firefly optimization algorithm is used to optimize the thermosiphon heat pipe's operational conditions. As inputs to the firefly algorithm, the filling ratio, nanofluid concentration and panel angle are considered in this step. The proposed machine learning model is trained as a cost function of the colonial competition algorithm, and its objective is to maximize the photovoltaic panel's output power. After optimization with the firefly algorithm, the optimal filling ratio, nanofluid concentration and panel angle are calculated. Due to their uncontrollability, the values of solar radiation and air temperature are presumed to be constant in the optimization algorithm. It is estimated that the minimum quantity of solar radiation is ∼1.5 kW/m^2, and the average air temperature is ∼24°C. Table 2 displays the optimized results of the firefly algorithm.

The subsequent analysis assesses the optimized outcomes of nanofluid concentration in comparison to other concentrations and compares it to water as a coolant in terms of the efficiency of solar panels for power generation.

3.3. Comparison results of different density and water

The thermal efficiency of the photovoltaic heat collector was compared to the thermal efficiency obtained from tests. The tests were conducted using water, nanofluid at concentrations of 0.5% and 1% and an optimal concentration of 0.86% by mass for cooling the system (Figure 6a). The thermal efficiency is regarded as the primary criterion for assessing the heat-capturing capability of the photovoltaic collector. The higher thermal efficiency of the system when using a nanofluid with a higher concentration, as opposed to a nanofluid with a lower concentration, is primarily attributed to its superior thermal conductivity coefficient.

Figure 6b illustrates the electrical efficiency of the photovoltaic collector using nanofluid different concentrations, as well as water as the working fluid. The tests conducted on the photovoltaic/thermal panel involved measuring both the thermal efficiency parameters and the electrical parameters of the system using two significant devices. These measurements were taken into account while considering the relationship between the parameters. The electrical efficiency of the system has been calculated for both modes of using water and nanofluid, taking

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

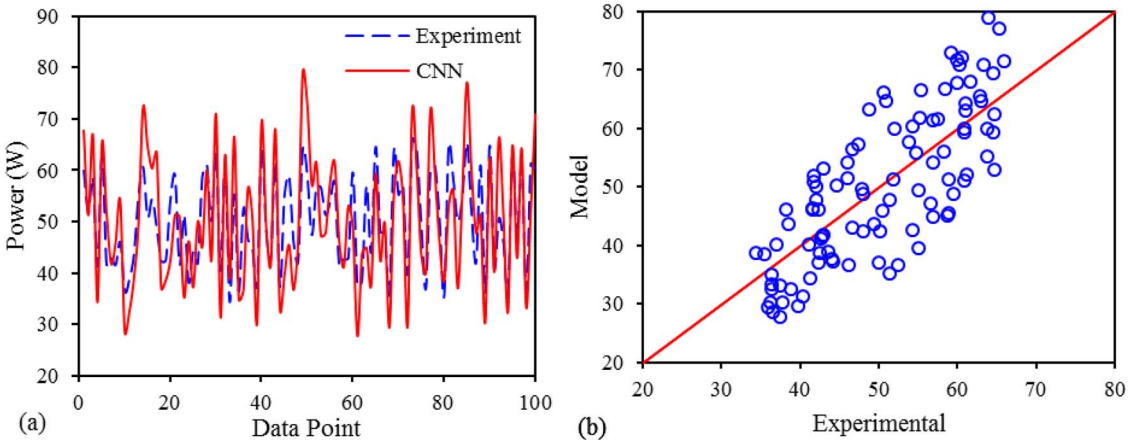


Figure 4. *A comparison between photovoltaic output power results of classic CNN model prediction and experimental data.*

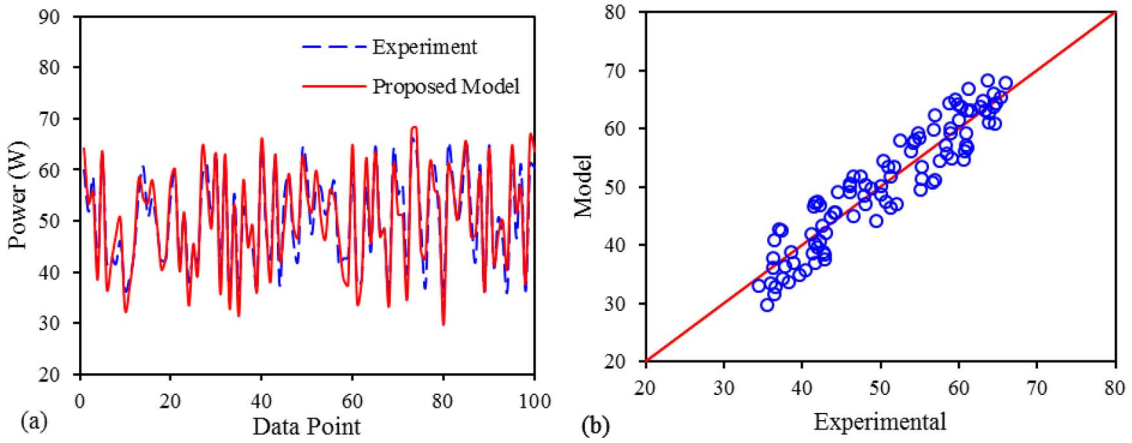


Figure 5. *A comparison between photovoltaic output power results of proposed model prediction and experimental data.*

Table 1. *Model performance analysis*

Model	RMSE (W)					
	Under 50 W		Upper 50 W		Overall	
	Train	Test	Train	Test	Train	Test
Classic CNN model	3.91	4.81	8.82	9.02	7.91	8.29
Proposed model	3.43	3.95	3.81	4.17	3.56	4.05

into account the specifications of the photovoltaic panel and the mentioned mass flows.

During all tests, the electrical efficiency initially reaches its peak in the morning and subsequently undergoes a gradual decline until it reaches its minimum value. The electrical efficiency experiences a subsequent increase and returns to a value near its original level at the start of the day.

The decline in electrical efficiency is primarily attributed to the significant influence of temperature on electrical efficiency. The lower ambient temperature in the early morning results in a decrease in the surface temperature of the cells, leading to the highest level of electrical efficiency. At noon, when the temperature and radiation intensity rise, the surface temperature of the cells increases, causing the electrical efficiency to reach its minimum value. Nevertheless, the electrical efficiency reaches its peak when nanofluid is utilized at the ideal concentration.

Figure 6c displays the exergy efficiency associated with the photovoltaic collector in scenarios where the system is cooled using either water or nanofluid. Given the superior quality of electrical energy in comparison to thermal energy, the device's exergy efficiency is heavily influenced by electrical efficiency.

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

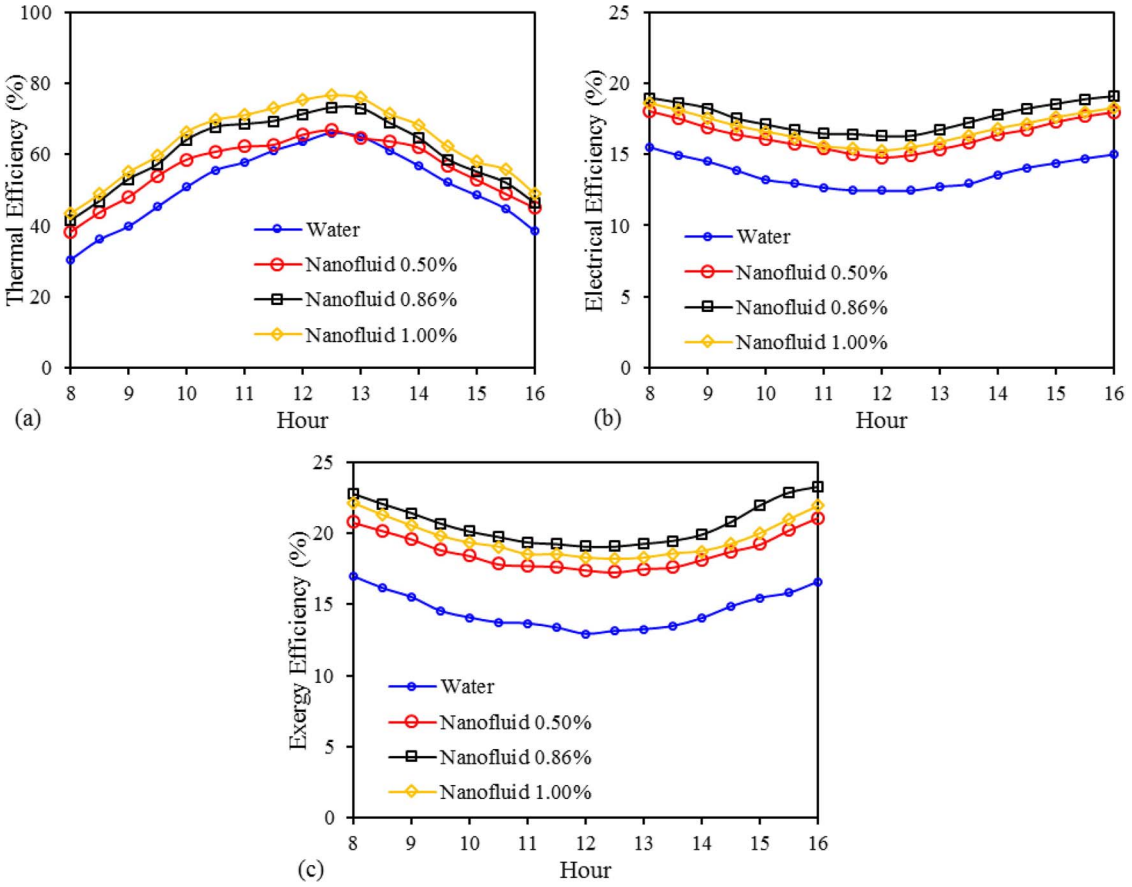


Figure 6. *A comparison between photovoltaic (a) thermal, (b) electrical and (c) exergy efficiency.*

Table 2. *Firefly optimization result*

Angle (°)	Filling ratio (%)	Nanofluid density (%)	Maximum power (W)
15	29	0.86	59.16

Furthermore, the behavior of exergy efficiency closely resembles that of electrical efficiency. The exergy efficiency is the highest for nanofluid at its optimal percentage, whereas cooling with water exhibits the lowest exergy efficiency.

4. Conclusion

Photovoltaic panels are a highly promising technology for generating solar energy, especially as the demand for renewable energy sources continues to increase. Efficient cooling is essential for achieving optimal performance and ensuring the longevity of these panels. Thermal effects can diminish the efficiency of photovoltaic panels and lead to permanent harm. Photovoltaic panel heat is typically regulated through the utilization of air and water cooling methods. The methods frequently encounter challenges related to efficiency and cost-effectiveness. In recent years, the cooling of photovoltaic panels has been enhanced by the implementation of advanced technologies such as heat pipes and nanofluids. Heat pipes are an innovative solution for dissipating heat in photovoltaic panels due to their exceptional heat transfer capabilities. Heat pipes employ the phenomenon of phase change in a working fluid to effectively transport heat from localized high-temperature regions on the surface of a panel to cooler areas, thereby enhancing heat dissipation. Heat pipes provide passive and reliable cooling for photovoltaic systems by utilizing evaporation and condensation processes. Utilizing nanofluids in heat pipes can enhance the efficiency of cooling photovoltaic panels. Nanofluids, which are created by dispersing nanoparticles in base fluids, possess distinctive thermal characteristics that enhance the process of heat transfer. Nanoparticles such as carbon nanotubes or metallic nanoparticles, when dispersed, enhance the thermal conductivity of the base fluid, thereby enhancing the dissipation of heat. This enhances the heat management and performance of photovoltaic panels. In order to enhance the cooling of photovoltaic panels, it is necessary to have a comprehensive understanding of the intricate interplay among heat pipes, nanofluids and the panel itself. The utilization of machine learning and optimization models for the analysis and enhancement of the cooling system is highly effective. Machine

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

learning models can be integrated with optimization techniques to ascertain the most favorable operating conditions for a heat pipe-nanofluid cooling system used in cooling photovoltaic panels. The optimization algorithm assumes constant solar radiation and air temperature values due to their uncontrollable nature. The firefly optimization algorithm is employed to optimize the operational parameters of the thermosiphon heat pipe. The algorithm takes into account the inputs of filling ratio, nanofluid concentration and panel angle. The objective of this study is to investigate the use of heat pipes and nanofluids to cool photovoltaic panels by employing hybrid machine learning and optimization models. The research aims to improve the understanding of the cooling process and develop strategies to optimize the cooling performance and power of photovoltaic panels by incorporating advanced technologies. The proposed model aims to enhance renewable energy technologies and establish a foundation for more effective and environmentally friendly photovoltaic systems. The main results of the study are summarized as follows:

- The forecasted outcomes of a CNN were compared to the actual values obtained from laboratory experiments. The RMSE index of 8.29 W indicates that the CNN model's predictions align with the experimental results, but it struggles to accurately forecast high values of photovoltaic panel output power. A scatter plot depicting the dispersion of model and laboratory results further confirms the lack of accuracy in estimating high power outputs of the photovoltaic panel.
- The proposed model, incorporating a CNN model and the firefly optimization algorithm, shows improved performance in accurately anticipating high power values of the solar panel. The RMSE index is reported as 4.05 W.
- The proposed model outperforms the classical model for values above 50 W. The suggested model achieves an RMSE of 3.95 W, while the classical model achieves 4.81 W for values under 50 W.
- The proposed machine learning model is trained using the colonial competition algorithm as a cost function, aiming to maximize the photovoltaic panel's output power.
- After optimization with the firefly algorithm, the optimal values for the filling ratio, nanofluid concentration and panel angle are calculated.
- A higher thermal efficiency in the photovoltaic collector when using nanofluid with a concentration of 1% by mass is attributed to its superior thermal conductivity coefficient.
- The electrical efficiency of the photovoltaic collector shows a diurnal pattern, peaking in the morning, declining at noon due to temperature and radiation and recovering by the start of the day. Optimal electrical efficiency is achieved with nanofluid at a concentration of 0.86%.
- Exergy efficiency closely shows electrical efficiency, with nanofluid at the optimal percentage leading to the highest efficiency, while cooling with water results in the lowest exergy efficiency.

Author contributions

Aysar Abdulrahman (Conceptualization [equal], Data curation [equal], Formal analysis [equal], Investigation [equal], Methodology [equal], Resources [equal], Software [equal], Validation [equal], Visualization [equal], Writing—original draft [equal]).

REFERENCES

[1] Olawuyi D. Sustainable development and the water-energy-food nexus: Legal challenges and emerging solutions. *Environ Sci Pol* 2020;**103**:1–9. https://doi.org/10.1016/j.envsci.2019.10.009.

[2] Naqvi SAH, Taner T, Ozkaymak M. *et al.* Hydrogen production through alkaline electrolyzers: a techno-economic and enviro-economic analysis. *Chem Eng Technol* 2023;**46**:474–81. https://doi.org/10.1002/ceat.202200234.

[3] Taner T. The novel and innovative design with using H2 fuel of PEM fuel cell: efficiency of thermodynamic analyze. *Fuel* 2021;**302**:121109. https://doi.org/10.1016/j.fuel.2021.121109.

[4] Alaiwi Y, Taha A, ALjumaili Y. Enhancement of the polycrystalline solar panel performance using a heatsink cooling system with PCM. *Int J Eng Artif Intell* 2023;**4**:24–34. https://doi.org/10.55923/JO.IJEAL.4.1.103.

[5] Cuce PM, Saxena A, Cuce E. *et al.* Applications of solar PV tree systems with different design aspects and performance assessment. *Int J Low Carbon Technol* 2022;**17**:266–78. https://doi.org/10.1093/ijlct/ctac004.

[6] Cui Y, Theo E, Gurler T. *et al.* A comprehensive review on renewable and sustainable heating systems for poultry farming. *Int J Low Carbon Technol* 2020;**15**:121–42. https://doi.org/10.1093/ijlct/ctz048.

[7] Jasim MA, Ahmed OK, Alaiwi Y. Performance of solar stills integrated with PV/Thermal solar collectors: a review. *J Renew Energy* 2023;**4**:97–111. https://doi.org/10.56286/ntujre.v4i1.456.

[8] Mutar WM, Alaiwi Y. Experimental investigation of thermal performance of single pass solar collector using high porosity metal foams. *Case Stud Therm Eng* 2023;**45**:102879. https://doi.org/10.1016/j.csite.2023.102879.

[9] Soomar AM, Hakeem A, Messaoudi M. *et al.* Solar photovoltaic energy optimization and challenges. *Front Energy Res* 2022;**10**:879985. https://doi.org/10.3389/fenrg.2022.879985.

[10] AlMallahi MN, El Haj Assad, AlShihabi S. *et al.* Multi-criteria decision-making approach for the selection of cleaning method of solar PV panels in United Arab Emirates based on sustainability perspective. *Inte J Low Carbon Technol* 2022;**17**:380–93. https://doi.org/10.1093/ijlct/ctac010.

[11] Yanto DTP, Akhmadeev R, Hamad HS. *et al.* Development and investigation of a pollutants emission reduction process from a coal-gasification power plant integrated with fuel cell and solar energy. *Int J Low Carbon Technol* 2023;**18**:1120–33. https://doi.org/10.1093/ijlct/ctad093.

[12] Menni Y, Chamkha AJ, Ameur H. *et al.* Hydrodynamic behavior in solar oil heat exchanger ducts fitted with staggered baffles and fins. *J Appl Comput Mech* 2022;**8**:774–90. https://doi.org/10.22055/jacm.2020.32468.2021.

[13] Abad SSAMK, Javidan P, Baghdadi M. *et al.* Green synthesis of Pd@biochar using the extract and biochar of corn-husk wastes for electrochemical Cr (VI) reduction in plating wastewater. *J Environ Chem Eng* 2023;**11**:109911. https://doi.org/10.1016/j.jece.2023.109911.

[14] Javidan P, Baghdadi M, Torabian A. *et al.* A tailored metal–organic framework applicable at natural pH for the removal of 17α-ethinylestradiol from surface water. *Desalin Water Treat* 2022;**264**:259–69. https://doi.org/10.5004/dwt.2022.28563.

[15] Olia H, Torabi M, Bahiraei M. *et al.* Application of nanofluids in thermal performance enhancement of parabolic trough solar collector: state-of-the-art. *Appl Sci* 2019;**9**:463. https://doi.org/10.3390/app9030463.

[16] Ramezanizadeh M, Alhuyi Nazari M, Ahmadi MH. *et al.* Application of nanofluids in thermosyphons: a review. *J Mol Liq* 2018;**272**:395–402. https://doi.org/10.1016/j.molliq.2018.09.101.

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

[17] Jasim MN, Alaiwi Y. Study the effect of adding heat exchanger on the refrigeration system performance. *Diyala J Eng Sci* 2023;**16**:1–15. https://doi.org/10.24237/djes.2023.16201.

[18] Choi SUS Eastman JA. Enhancing thermal conductivity of fluids with nanoparticles. American Society of Mechanical Engineers, Fluids Engineering Division (Publication) FED, 1995.

[19] Liao J, Zhang A, Qing S. *et al.* Investigation on the aggregation structure of nanoparticle on the thermal conductivity of nanofluids by molecular dynamic simulations. *Powder Technol* 2022;**395**:584–91. https://doi.org/10.1016/j.powtec.2021.10.007.

[20] Sahin F, Genc O, Gökcek M. *et al.* An experimental and new study on thermal conductivity and zeta potential of Fe3O4/water nanofluid: machine learning modeling and proposing a new correlation. *Powder Technol* 2023;**420**:118388. https://doi.org/10.1016/j.powtec.2023.118388.

[21] He D, Li C, Chen Y. *et al.* Prediction of thermal conductivity of hybrid nanofluids based on deep forest model. *Heat Transf Res* 2022;**53**:55–71. https://doi.org/10.1615/HeatTransRes.2022041841.

[22] Miao T-T, Xia Y, Chen D-S. *et al.* Significant enhancement of thermal conductivity in toluene-based nanofluids employing highly dispersed and concentrated ZrO2 nanodots. *Int J Therm Sci* 2022;**182**:107819. https://doi.org/10.1016/j.ijthermalsci.2022.107819.

[23] Chen W, Zhai Y, Guo W. *et al.* A molecular dynamic simulation of the influence of linear aggregations on heat flux direction on the thermal conductivity of nanofluids. *Powder Technol* 2023;**413**:118052. https://doi.org/10.1016/j.powtec.2022.118052.

[24] Guan H, Su Q, Wang R. *et al.* Why can hybrid nanofluid improve thermal conductivity more? A molecular dynamics simulation. *J Mol Liq* 2023;**372**:121178. https://doi.org/10.1016/j.molliq.2022.121178.

[25] Abdullatif Alshuhail L, Shaik F, Syam Sundar L. Thermal efficiency enhancement of mono and hybrid nanofluids in solar thermal applications – a review. *Alex Eng J* 2023;**68**:365–404. https://doi.org/10.1016/j.aej.2023.01.043.

[26] Younes H, Mao M, Sohel Murshed SM. *et al.* Nanofluids: key parameters to enhance thermal conductivity and its applications. *Appl Therm Eng* 2022;**207**:118202. https://doi.org/10.1016/j.applthermaleng.2022.118202.

[27] Yasmin H, Giwa SO, Noor S. *et al.* Thermal conductivity enhancement of metal oxide nanofluids: a critical review. *Nanomaterials* 2023;**13**:597. https://doi.org/10.3390/nano13030597.

[28] Lin H, Jian Q, Bai X. *et al.* Recent advances in thermal conductivity and thermal applications of graphene and its derivatives nanofluids. *Appl Therm Eng* 2023;**218**:119176. https://doi.org/10.1016/j.applthermaleng.2022.119176.

[29] Mun H-S, Dilawar MA, Mahfuz S. *et al.* Effects of a combined geothermal and solar heating system as a renewable energy source in a pig house and estimation of energy consumption using artificial intelligence-based prediction model. *Animals* 2022;**12**:2860. https://doi.org/10.3390/ani12202860.

[30] Liu Y, Hu D, Luo X. *et al.* Design optimization of centralized–decentralized hybrid solar heating system based on building clustering. *Energies* 2022;**15**:1019. https://doi.org/10.3390/en15031019.

[31] Yu T, Wang D, Zhao X. *et al.* Experimental and numerical study of an active solar heating system with soil heat storage for greenhouses in cold climate zones. *Buildings* 2022;**12**:405. https://doi.org/10.3390/buildings12040405.

[32] Mehdaoui F, Hazami M, Messaouda A. *et al.* Performance analysis of two types of solar heating systems used in buildings under typical North-African climate (Tunisia). *Appl Therm Eng* 2020;**165**:114203. https://doi.org/10.1016/j.applthermaleng.2019.114203.

[33] Xia T, Li Y, Sun Z. *et al.* Performance study of an active solar water curtain heating system for Chinese solar greenhouse heating in high latitudes regions. *Appl Energy* 2023;**332**:120548. https://doi.org/10.1016/j.apenergy.2022.120548.

[34] Razak A, Irwan Y, Leow WZ. *et al.* Investigation of the effect temperature on photovoltaic (PV) panel output performance. *Int J Adv Sci Eng Info Technol* 2016;**6**:682. https://doi.org/10.18517/ijaseit.6.5.938.

[35] El Hammoumi A, Chtita S, Motahhir S. *et al.* Solar PV energy: from material to use, and the most commonly used techniques to maximize the power output of PV systems: a focus on solar trackers and floating solar panels. *Energy Rep* 2022;**8**:11992–2010. https://doi.org/10.1016/j.egyr.2022.09.054.

[36] Shah IA, Bilal S, Akgül A. *et al.* Significance of cold cylinder in heat control in power law fluid enclosed in isosceles triangular cavity generated by natural convection: a computational approach. *Alex Eng J* 2022;**61**:7277–90. https://doi.org/10.1016/j.aej.2021.12.071.

[37] Kargaran M, Goshayeshi HR, Pourpasha H. *et al.* An extensive review on the latest developments of using oscillating heat pipe on cooling of photovoltaic thermal system. *Therm Sci Eng Prog* 2022;**36**:101489. https://doi.org/10.1016/j.tsep.2022.101489.

[38] Papari MM, Yousefi F, Moghadasi J. *et al.* Modeling thermal conductivity augmentation of nanofluids using diffusion neural networks. *Int J Therm Sci* 2011;**50**:44–52. https://doi.org/10.1016/j.ijthermalsci.2010.09.006.

[39] Hojjat M, Etemad SG, Bagheri R. *et al.* Thermal conductivity of non-Newtonian nanofluids: experimental data and modeling using neural network. *Int J Heat Mass Transf* 2011;**54**:1017–23. https://doi.org/10.1016/j.ijheatmasstransfer.2010.11.039.

[40] Longo GA, Zilio C, Ceseracciu E. *et al.* Application of artificial neural network (ANN) for the prediction of thermal conductivity of oxide–water nanofluids. *Nano Energy* 2012;**1**:290–6. https://doi.org/10.1016/j.nanoen.2011.11.007.

[41] Hemmat Esfe M, Alirezaie A, Rejvani M. An applicable study on the thermal conductivity of SWCNT-MgO hybrid nanofluid and price-performance analysis for energy management. *Appl Therm Eng* 2017;**111**:1202–10. https://doi.org/10.1016/j.applthermaleng.2016.09.091.

[42] Afrand M, Toghraie D, Sina N. Experimental study on thermal conductivity of water-based Fe3O4 nanofluid: development of a new correlation and modeled by artificial neural network. *Int Commun Heat Mass Transfer* 2016;**75**:262–9. https://doi.org/10.1016/j.icheatmasstransfer.2016.04.023.

[43] Venkatakrishnan GR, Rengaraj R, Tamilselvi S. *et al.* Detection, location, and diagnosis of different faults in large solar PV system—a review. *Int J Low Carbon Technol* 2023;**18**:659–74. https://doi.org/10.1093/ijlct/ctad018.

[44] Said Z, Sharma P, Elavarasan RM. *et al.* Exploring the specific heat capacity of water-based hybrid nanofluids for solar energy applications: A comparative evaluation of modern ensemble machine learning techniques. *J Energy Storage* 2022;**54**:105230. https://doi.org/10.1016/j.est.2022.105230.

[45] Madvar MD, Nazari MA, Arjmand JT. *et al.* Analysis of stakeholder roles and the challenges of solar energy utilization in Iran. *Int J Low Carbon Technol* 2018;**13**:438–51. https://doi.org/10.1093/ijlct/cty044.

[46] Nejatian N, Yavary Nia M, Yousefyani H. *et al.* The improvement of wavelet-based multilinear regression for suspended sediment load modeling by considering the physiographic characteristics of the watershed. *Water Sci Technol* 2023;**87**:1791–802. https://doi.org/10.2166/wst.2023.089.

[47] Chen TC, Najat Rashid Z, Theruvil Sayed B. *et al.* Evaluation of hybrid soft computing model's performance in estimating wave height. *Adv Civi Eng* 2023;**2023**:1–13. https://doi.org/10.1155/2023/8272566.

[48] Nourani V, Molajou A, Najafi H. *et al.* Emotional ANN (EANN): a new generation of neural networks for hydrological modeling in IoT. *Artificial intelligence in IoT* 2019;45–61.

[49] Sharghi E, Nourani V, Najafi H. *et al.* Emotional ANN (EANN) and Wavelet-ANN (WANN) approaches for Markovian and seasonal based modeling of rainfall-runoff process. *Water Resour Manag* 2018;**32**:3441–56. https://doi.org/10.1007/s11269-018-2000-y.

[50] Yamashita R, Nishio M, Do RKG. *et al.* Convolutional neural networks: an overview and application in radiology. *Insights Imaging* 2018;**9**:611–29. https://doi.org/10.1007/s13244-018-0639-9.

[51] Gu J, Wang Z, Kuen J. *et al.* Recent advances in convolutional neural networks. *Pattern Recogn* 2018;**77**:354–77. https://doi.org/10.1016/j.patcog.2017.10.013.

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026

[52] Lyu S, Liu J. Convolutional recurrent neural networks for text classification. *J Database Manag* 2021;**32**:65–82. https://doi.org/10.4018/JDM.2021100105.
[53] Toumia I, Hassine AB. Impact of input data structure on convolutional neural network energy prediction model. *Int J Adv Comput Sci Appl* 2022;**13**:748–57. https://doi.org/10.14569/IJACSA.2022.0131089.
[54] Chen SY-C, Wei T-C, Zhang C. *et al.* Quantum convolutional neural networks for high energy physics data analysis. *Phys Rev Res* 2022;**4**:013231. https://doi.org/10.1103/PhysRevResearch.4.013231.
[55] Chin W, Zhang Q, Jiang T. Low-complexity neuron for fixed-point artificial neural networks with ReLU activation function in energy-constrained wireless applications. *IET Commun* 2021;**15**:917–23. https://doi.org/10.1049/cmu2.12129.
[56] Torres AD, Yan H, Aboutalebi AH. *et al.* 2018. Patient facial emotion recognition and sentiment analysis using secure cloud with hardware acceleration. In *Computational Intelligence for Multimedia Big Data on the Cloud with Engineering Applications*. Elsevier. 61–89.
[57] Chen J, Liu Z, Wang H. *et al.* Automatic defect detection of fasteners on the catenary support device using deep convolutional neural network. *IEEE Trans Instrum Meas* 2018;**67**:257–69. https://doi.org/10.1109/TIM.2017.2775345.
[58] Kaselimi M, Protopapadakis E, Voulodimos A. *et al.* Multi-channel recurrent convolutional neural networks for energy disaggregation. *IEEE Access* 2019;**7**:81047–56. https://doi.org/10.1109/ACCESS.2019.2923742.
[59] Behera MP, Sarangi A, Mishra D. *et al.* Optimizing multi-layer perceptron using variable step size firefly optimization algorithm for diabetes data classification. *Int J Online Biomed Eng* 2023;**19**:124–39. https://doi.org/10.3991/ijoe.v19i04.36543.
[60] Dai W, Liang L, Zhang B. Firefly optimization algorithm for the prediction of uplift due to high-pressure jet grouting. *Adv Civ Eng* 2020;**2020**:1–12. https://doi.org/10.1155/2020/8833784.
[61] Taleizadeh AA. Stochastic multi-objectives supply chain optimization with forecasting partial backordering rate: a novel hybrid method of meta goal programming and evolutionary algorithms. *Asia-Pac J Oper Res* 2017;**34**:1750021. https://doi.org/10.1142/S021759591750021X.
[62] Lin S, Broadbent J, McGlen R. Numerical study of heat pipe application in heat recovery systems. *Appl Therm Eng* 2005;**25**:127–33. https://doi.org/10.1016/j.applthermaleng.2004.02.012.

Downloaded from academic.oup.com/ijlct/article/doi/10.1093/ijlct/ctae047/7656033 by guest on 19 October 2026